

EEE3095/6S Class Test Solutions

October 2024

Duration: 75 minutes

Total: 50 marks

Empl ID: _____

Instructions:

- This is a closed-book test.
- There are **two** Sections (A and B) to this test.
- You must use a pen to complete the test – pencil will only be marked for drawings. Values indicated on drawings must be written in pen.
- Please show all your working clearly – your method and approach matter.
- Keep all your answers within the allocated blocks; anything outside of these blocks will not be marked.

Question	Available Marks	Grade
1	12	
2	8	
3	10	
4	10	
5	10	
Total:	50	

Section A: Multiple Choice

Empl ID:

Question:	1	2	Total
Points:	12	8	20
Score:			

Question 1 [12 marks]

- a. If the Serial Peripheral Interface (SPI) protocol is configured to operate in SPI Mode 1, the logic level of the clock signal is: (2)
- ☒ **low while idling**
☐ high while idling
☐ always low
☐ always alternating
- b. Loop Unrolling (LUR) is a common optimisation strategy that... (2)
- ☐ forces the processor to ignore loop condition checks to reduce overhead
☐ parallelises iterations of a loop to improve runtime efficiency
☐ removes unnecessary loop statements
☒ **reduces the number of iterations in a loop to reduce overhead**
- c. Serial Peripheral Interface (SPI) and Inter-Integrated Circuit (I2C) each use a different means of selecting a slave device with which the master communicates. From the following options, choose the most accurate descriptions. (2)
- ☐ SPI and I2C both use a Slave Select line to initiate communication with a specific slave device.
☒ **With I2C, the master transmits a 7-bit address to indicate its slave device selection, whereas SPI uses a dedicated Slave Select line.**
☐ With SPI, the master transmits a 7-bit address to indicate its slave device selection, whereas I2C uses a dedicated Slave Select line.
☐ SPI and I2C both use 7-bit addressing to indicate slave device selection.
- d. The STM32F051C6's non-volatile memory, also known as Flash memory, comprises the memory address space between... (2)
- ☐ 0x20000000 and 0x20001FFF
☒ **0x08000000 and 0x08007FFF**
☐ 0x20000000 and 0x480017FF
☐ 0x08000000 and 0x480017FF
- e. To facilitate RS-232-compliant communications with our STM board, the MAX232 chip converts our voltages from 0 V and 3.3 V, respectively, to... (2)
- ☒ **+12 V and -12 V**
☐ -12 V and +12 V
☐ 0 V and 12 V
☐ -12 V and 0 V
- f. For which reason below is RISC being shown to be more popular than CISC for embedded systems development? (2)

- ☐ It does more processing in a clock cycle
- ☒ **It is more cost-effective and more streamlined**
- ☐ It gives better real-time performance
- ☐ It is easier to use and has more debugging support

Question 2 [8 marks]

Select either True or False for each of the following statements.

- a. The 7-bit data word 0b1110111 is to be transmitted with a parity bit appended; if using odd parity, the final 8-bit data word that would be transmitted is 0b11101111. (1)
- ☒ **True** ☐ False
- b. RS-422 allows for full-duplex communication between one transmitter and up to 32 receivers. (1)
- ☐ True ☒ **False**
- c. The FU (Function Unit) is considered to be a part of the ALU (Arithmetic Logic Unit). (1)
- ☐ True ☒ **False**
- d. An I2C slave sends a NACK bit when it is unable to receive/transmit because it is busy and thus not ready to start communication with the master. (1)
- ☒ **True** ☐ False
- e. A timer works by having the CNT register count up to the value in the PSC, starting from 0; then, when $CNT = PSC$, an overflow occurs and CNT is reset to 0. (1)
- ☐ True ☒ **False**
- f. The memory address 0x08000004 corresponds to the location of the Reset Vector. True or false? (1)
- ☒ **True** ☐ False
- g. In the general 3-stage CPU example discussed in class, the sequencing of the stages involved write-back as the first stage that gets executed in the sequence of actuating an instruction. (1)
- ☐ True ☒ **False**
- h. Thumb instructions can not only save space, but can also help increase overall program execution speeds compared to AArch32. (1)
- ☒ **True** ☐ False

Solution: Indeed this is so, but not necessarily always the case. Thumb could actually end up having slower execution compared to AArch32, even if less code, due to things like very little space for constants and fewer operations that can be used.

Section B: Full Questions

Empl ID:

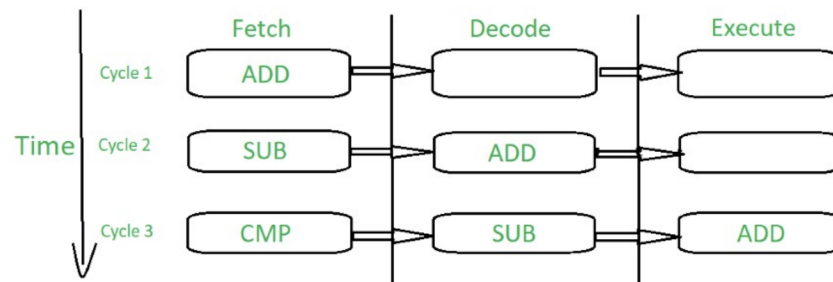
Question:	3	4	5	Total
Points:	10	10	10	30
Score:				

Question 3 [10 marks]

- a. The ARM Cortex-M0 is said to have a three-stage pipeline. Draw a diagram to illustrate what this means, and briefly explain how it works.

(6)

Solution: Figure below. A three-stage pipeline means that the processor runs instructions one-by-one using a pipelined system in three steps. So, firstly, it fetches or loads an instruction from memory; secondly, it then decodes/identifies the instruction; and then thirdly, it executes/processes the instruction. But the processor actually parallelises these three steps, so while one instructions is being decoded, the next instruction is being fetched; and while one is being executed, the next is being decoded and another one is being fetched, and so on.



- b. Describe the terms "execution environment" and "runtime environment", and explain the relationship between them.

(4)

Solution: The runtime environment is a computer-based environment where a developed program is modelled to run, addressing various issues such as how the program starts, how functions are called, memory management, and more.

The execution environment refers to those components that are used together with the application's code to make a complete computing system, such as the processors, networks, operating systems and so on.

The *runtime* environment can thus be considered a large part of the *execution* environment, and an execution environment could comprise multiple runtime environments.

Question 4 [10 marks]

Imagine that you are developing firmware for an embedded system using a 32-bit microcontroller. The system utilises memory-mapped I/O to interface with an 8-bit ADC and a 32-bit timer.

The ADC provides the latest analog sample value at address 0xFFFF1000. The timer increments every 10 μs and is accessible at address 0xFFFF4000. Do not make any assumptions about the value at which the timer starts.

Write a C code snippet that determines the time it takes for the analog input signal to transition from a *trigger on* value – which for this program is set at a constant value of 200 – to a value below a *trigger off* value – in this program, use a constant value of 100.

Requirements:

Use pointers to access the memory-mapped I/O addresses.

Calculate the time difference in microseconds.

Store the calculated time difference in a variable named *time_difference*.

Example run case:

Here is an example case of running the program to help explain its operation. Consider that the analog input reaches 200 or above at time $t = 500 \mu\text{s}$, and then falls below 100 at $t = 5000 \mu\text{s}$; in this case, the *time_difference* should be 4500 μs .

Solution:

```
#include <stdint.h>

volatile uint8_t *adc_ptr    = (uint8_t *)0xFFFF1000; // Pointer to ADC
volatile uint32_t *timer_ptr = (uint32_t *)0xFFFF4000; // Pointer to timer

uint32_t time_difference;
uint32_t start_time, end_time;

int main() {
    // Wait for the signal to go above 200
    while (*adc_ptr < 200);
    start_time = *timer_ptr;

    // Wait for the signal to go below 100
    while (*adc_ptr >= 100) {
        end_time = *timer_ptr;
        if (end_time < start_time) // Adjust for any timer wrap around:
            end_time += 0xFFFFFFFF; // Adjust end_time to account for wrap
    }

    // Calculate the time difference in microseconds
    time_difference = (end_time - start_time) * 10;

    // ... further code here, e.g. sending time difference ...
    return 0;
}
```

Question 5 [10 marks]

- a. Describe one benefit and one drawback to using RS-232 over I2C for embedded communications.

(2)

Solution:**Benefits of RS-232:**

- It is full duplex whereas I2C is half-duplex
- It supports longer distances than I2C
- Higher voltage levels mean better resilience to noise (and longer transmission distances)

Drawbacks of RS-232:

- It is suitable for system-to-system communications, but it is not suitable for chip-to-chip or chip-to-sensor device communications
- Supports lower speed for long distances whereas I2C supports faster speed for shorter distances
- It can only be used in a single-master and single-slave configuration, whereas I2C supports multiple masters and multiple slaves
- It needs multiple wires for communication, whereas I2C needs only two wires
- It is unbalanced transmission

- b. When using SPI, is it necessary to have a common clock line between communicating devices? Why? And where would this signal be generated? (3)

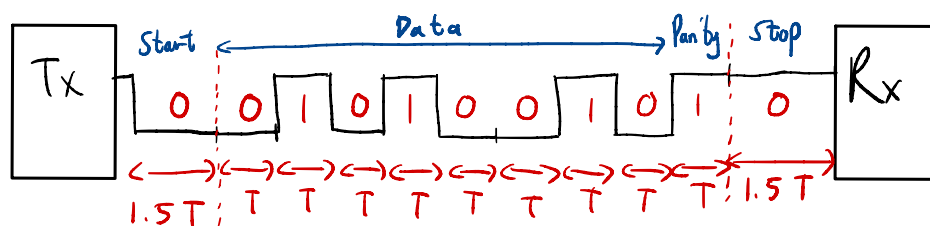
Solution: SPI is a *synchronous* protocol and thus requires a clock line (SCLK) that connects all the devices; all slave devices should be connected to the master via the common clock line, and this signal is generated directly by the master device. The clock thus ensures that the transmitting and receiving devices are synchronised and no start/stop bits are required.

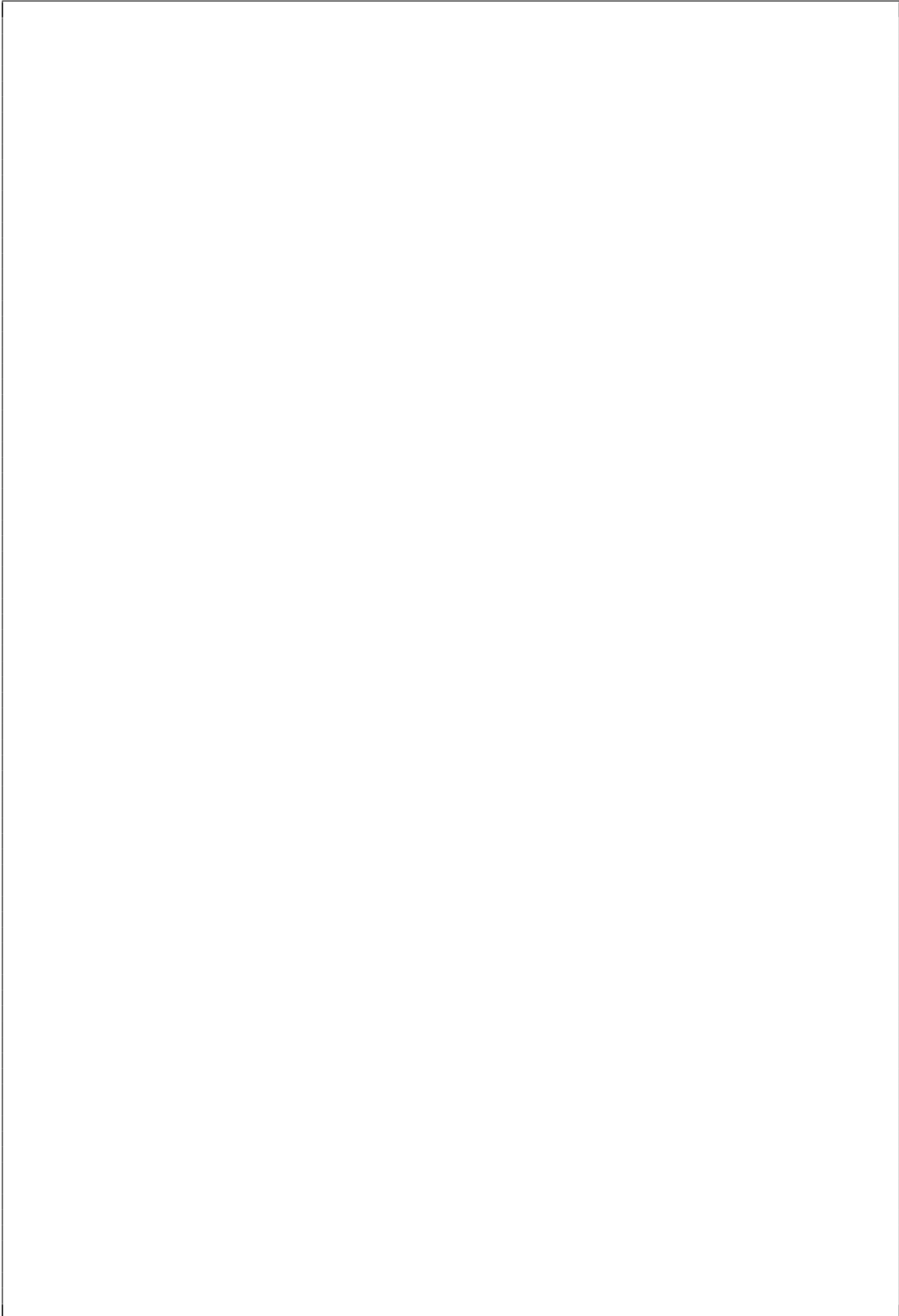
- c. You have been asked to use RS-232 to transmit the data byte 0x4A to a receiving device. Assuming that Little Endian and even parity are being used, draw an appropriate timing diagram to illustrate this communication process. (5)

Solution:

Something like the figure below; students could use logic levels or voltage levels (essentially "inverted"), but they must have an actual **timing diagram** to represent the information [1], and the diagram should be adequately labelled [1].

Start/Stop bits (logic Low and High, respectively) must be included [1] as well as a parity bit – which should have a value of 1 due to even parity being used [1]. The transmitted data byte itself should be 0b01001010 (0x4A), but sent in "reverse" (lsb first) due to using Little Endian [1].





END OF TEST